Regime-Dependent Delta Hedging with SVI-Calibrated Volatility Surfaces on SPX Index Options

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THE GAP

SVI is the industry standard for volatility surface calibration. The literature validates fitting quality (10–50 bps RMSE) but stops there. **No study has tested whether a better surface fit produces a better hedge.** We close that loop on 2,000 real SPX options across four VIX regimes.

THE EXPERIMENT

28.6M SPX options → **5.6M** filtered → **2,000** stratified (500 per regime)

5 vol inputs → BSM Δ (daily hedge to expiry)
4 VIX regimes → Low / Normal / High / Crisis
Decomposition → El Karoui et al. (1998)

THREE HYPOTHESES

H1: SVI reduces error vs. flat BSM — **REJECTED**
H2: Yang-Zhang beats Close-to-Close — **REJECTED**
H3: Optimal input is regime-dependent — **PARTIAL**

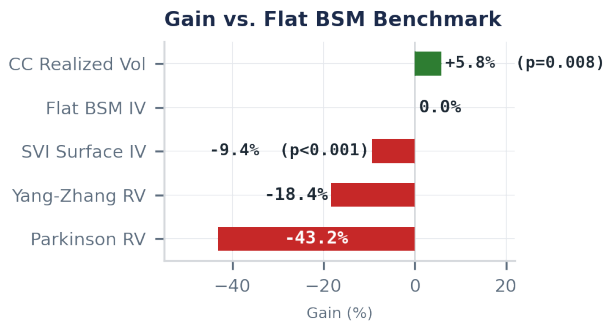
WHY DOES THE SMILE HURT?

1. **Calibration noise:** 5-param optimization → noisy Δ
2. **Interpolation:** calibrate every 5th day → stale smile
3. **Signal < noise** for most strikes—only OTM calls have steep enough slope to carry information

Parkinson is 5× more efficient than CC under GBM (Parkinson 1980), yet produces the worst hedging (−43.2%). Statistical efficiency ≠ hedging efficiency.

THE HORSE RACE

Gain vs. Flat BSM Benchmark



Hull & White (2017) report 25.7% Gain on SPX calls using minimum-variance delta—up to 42.1% for deep OTM. Our CC RV achieves +5.8% with a simpler method; the moneyness-conditional strategy targets the same OTM region where their gains are largest.

REGIME BREAKDOWN

Why H3 is "partial"

SVI & CC Gain by VIX Regime

SVI helps only in Normal/High VIX regimes

	Low	Normal	High	Crisis
BSM Std ($)	$18.4	$18.2	$20.6	$64.4
SVI Gain (%)	-8.8%	+0.8%	+3.7%	-2.3%
CC Gain (%)	-7.1%	-15.3%	-12.4%	-4.0%

THE TWIST

Best Vol Input by Moneyness × DTE

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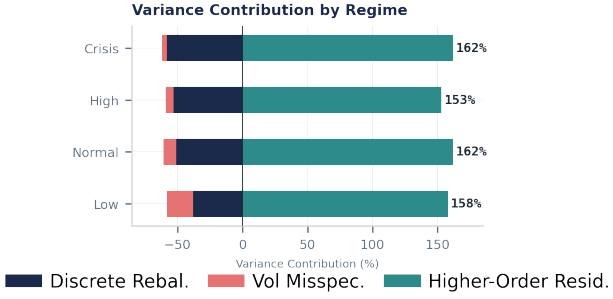
	Short	Medium	Long
OTM Call	SVI $23.0	SVI $25.6	YZ $29.8
ATM	BSM $24.4	YZ $24.0	YZ $23.4
OTM Put	BSM $42.3	CC $25.5	CC $23.0

Values = RMSE ($). Lowest wins each cell.

P&L ATTRIBUTION

Variance Contribution by Regime



KEY NUMBERS

Metric	Value
Sample	2,000 SPX options
Period	Jan 2019 – Dec 2024
SVI RMSE	19.5 bps median
Arb-free rate	68.6%
Best overall	CC RV (+5.8%)
Worst overall	Park (−43.2%)
Δ-Γ hedge	46% std reduction
Unit tests	80 (all passing)

PRACTITIONER TAKEAWAY

OTM Calls → SVI Surface IV (+6–12%)
ATM → Flat BSM IV
OTM Puts → CC Realized Vol (+21–48%)
VIX ≥ 35 → Focus on gamma management

Bottom line: No single vol input wins everywhere. A moneyness-conditional strategy outperforms any single approach.

Economic impact: For a $1B notional SPX book, 5.8% hedging error reduction ≈ 5–8% lower VaR, freeing risk capital.

Ruf & Wang (2022) show simple regressions match neural networks on real SPX—our regime-conditional approach targets the same channel with full interpretability.

The counterintuitive finding: the industry-standard volatility surface makes hedging worse. Ask me why.

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github.com/zaid282802/options-pricing-hedging

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Full 40-page thesis and Python code available upon request